IRWA Project – Part 1: Text processing and exploratory data analysis

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Repository: https://github.com/Vagaroto/RIAW-Presentaci-nFinal. The code is in `Part1/`, and the `README.md` explains how to run it and which file holds each number. Every table of this report names its source CSV file.

1. Data preparation

1.1 Text processing

Every indexed text field goes through `preprocess_text` (`DataPreparation.py`):

1. **Unicode normalization (NFKD) and accent removal**, so `Azúl` and `azul` are the same term.
2. **Lowercasing.**
3. **Joining `t-shirt`** (and `t shirt`) into `tshirt`.
4. **Tokenization and punctuation removal:** the text is split into runs of word characters, so punctuation acts as a separator. This also fixes missing spaces (`look.Designed`, `Comfort,Skin`).
5. **Stop-word removal** (NLTK English list).
6. **Stemming** (NLTK Snowball).

- **Extras beyond the statement:** steps 1 and 3.
- **Why step 3 is needed:**
  - The NLTK stop-word list contains the letter `t`, so `T-Shirt` would become `shirt`, a different product.
  - 12,247 of the 28,080 titles contain `T-Shirt` (`data_checks_results/preprocessing_checks.csv`).
  - `Sweatshirt` is one word and isn't affected.
- **Known limitation:** the NLTK list also removes `no` and `not`, so negations are lost (e.g. the value `Reversible = No`). This was not changed.

Original	Processed tokens
`Solid Women Multicolor Track Pants`	solid, women, multicolor, track, pant
`Typography Women V Neck Multicolor T-Shirt (Pack of 2)`	typographi, women, v, neck, multicolor, tshirt, pack, 2
`...rich combed cotton giving it a rich look.Designed for Comfort,Skin friendly fabric...`	…, rich, comb, cotton, give, rich, look, design, comfort, skin, friend, fabric, …

1.2 Stored object and returned fields

- **Each product is stored as `{pid, original, processed, numeric}`:**
  - `original`: the untouched record.
  - `processed`: one token list per field, plus the merged `text` that is indexed.
  - `numeric`: the values kept for ranking (section 3).
- **Retrieval uses `processed`, but results are always returned from `original`.** So `pid`, `title`, `description`, `brand`, `category`, `sub_category`, `product_details`, `seller`, `out_of_stock`, `selling_price`, `discount`, `actual_price`, `average_rating` and `url` are returned unchanged.
- **`pid` is never modified**, because the evaluation uses it.

2. Handling of `category`, `sub_category`, `brand`, `product_details` and `seller`

2.1 Decision and approaches considered

Decision:

- `title`, `description`, `sub_category`, `brand` and `product_details` are **merged into one indexed text field, without field weights**.
- `category` and `seller` are **not indexed**, but they're still returned with the results.
- Of `product_details` (a list of `{name: value}` pairs) only the values are kept: `Fit: Slim` gives `slim`. The names repeat in thousands of products and would only add noise.

Approach	Pros	Cons
One merged field (chosen)	Simple index and one ranking function; no weights to tune; a query term matches wherever it appears.	The origin of a term is lost. Long fields (`product_details`, 24 tokens on average) dilute short distinctive ones (`brand`, `sub_category`). Field importance can't be changed.
Separate fields in the inverted index	Each field can be searched or boosted alone; short fields aren't diluted.	Needs weights to combine the fields, and 40 labeled products and 2 queries are too few to estimate them. Sparse fields give no score (`description` is empty in 39.7% of the products, `brand` in 8.0%). Larger index.
Low-cardinality fields as filters or facets	Exact results for values like `sub_category`.	The query would have to be translated into a filter, which a free-text search doesn't do.

The evidence of section 2.3 is enough to decide which fields to use, but not how much each should weigh. That is why the merged field was chosen.

2.2 Distinctiveness of each field

Field	Empty	Distinct values	Most common value	Consequence
`category`	0.0%	4	Clothing and Accessories (96.6%)	Almost constant: doesn't separate products
`sub_category`	0.0%	24	Topwear (54.5%)	Few values: low distinctiveness alone
`brand`	8.0%	324	ECKO Unl (3.4%)	Distinctive: identifies a product line
`seller`	6.0%	534	RetailNet (5.0%)	Distinctive, but describes the vendor, not the product
`product_details`	0.0%	95.3% of products differ	–	Richest field: fit, sleeve, fabric, colour, gender, neck

Source: `field_selection_results/field_profile.csv`.

2.3 Evaluation with the validation labels

Method:

- The 40 labeled products (2 queries) were ranked with BM25 for each of the 127 combinations of fields, each combination merged into one text.
- Each combination was scored with the **AUC**: the probability that a relevant product is ranked above a non-relevant one (1.0 = perfect, 0.5 = random).

Field	AUC alone	Change in AUC when removed from the full set
`title`	0.600	−0.076
`description`	0.586	−0.064
`product_details`	0.868	−0.035
`category`, `sub_category`, `brand`, `seller`	0.500	0.000

Source: `ablation.csv`, `combinations.csv` (all combinations) and `rankings.csv` (position of every labeled product), in `field_selection_results/`.

- **The full set scores 0.862, and so does the chosen set.** The best combination is `product_details` alone, at 0.868. A difference that small is noise with 40 labels.
- **The `t-shirt` step didn't change any score.** Neither query contains `shirt`, so these labels can't show its effect.

2.4 Decision for each field

- **`title`, included:**
  - Removing it hurts the most (−0.076).
  - It's weak alone because every labeled product already matches the query in its title.
- **`description`, included:** removing it costs −0.064. It's empty in 39.7% of the products, so it can't be the only source of terms.
- **`product_details`, included:**
  - It's the strongest field (0.868 alone), because it carries what separates relevant from non-relevant products:
    - Query 1: `Sleeve = Full Sleeve` is relevant in 13 of 16 products, `Half Sleeve` in 0 of 4.
    - Query 2: `Ideal For = Men` is relevant in 7 of 9, `Ideal For = Women` in 3 of 11.
  - Source: `field_values.csv`.
- **`sub_category`, included:**
  - Its effect is 0.000, because no query word appears in it.
  - It is still related to relevance: for query 1, `Winter Wear` is relevant in 13 of 16 products, `Topwear` in 0 of 4.
  - It's short, and it lets queries such as "winter wear" match.

- **`brand`, included by reasoning:**
  - Neither query names a brand, so the labels can't judge it.
  - Brand names are distinctive, and users search by brand.
  - Its few tokens are diluted by the long fields (see also section 4.4).
- **`seller`, not indexed:**
  - Its effect is 0.000.
  - The same sellers appear on relevant and non-relevant products (`BioworldMerchandising`: 4 relevant of 8).
  - It describes the vendor, so a query word could match a seller by accident.
- **`category`, not indexed:** 96.6% of the products share one value, so its IDF is close to zero.
- **Limits of this evidence:**
  - Only 40 labels and 2 queries.
  - The labeled products look like the top results of an earlier search, so they all match the title, which understates `title`.
  - `category`, `sub_category`, `brand` and `seller` never contain a query word, so these four were decided by reasoning.
  - Some labels look noisy (women's jeans marked relevant for a "men" query).

3. Numeric fields (`out_of_stock`, `selling_price`, `discount`, `actual_price`, `average_rating`)

Decision: these fields are **not indexed as text terms**. They're converted into clean values in the `numeric` object, for the ranking stage. The original strings stay in `original` and are the ones returned.

Field	Original	Stored	Missing	Range
`selling_price`	`"921"`	float	2 products	99 to 7,999
`actual_price`	`"2,999"`	float (separator removed)	777 (2.8%)	150 to 12,999
`discount`	`"69% off"`	percentage as a float (`69.0`)	855 (3.0%)	1 to 87
`average_rating`	`"3.9"`	float, 1 to 5	2,261 (8.1%)	1.0 to 5.0
`out_of_stock`	boolean	boolean	none	1,644 out of stock (5.9%)

Source: `data_checks_results/numeric_fields.csv`.

- **Numbers as text lose their meaning.** `999` and `1000` would be unrelated terms, so the index couldn't tell that two prices are close, or answer "cheaper" or "better rated".
- **They describe the offer, not the product.** Neither reference query mentions a price or a rating, and a term like `69` could match an unrelated product by accident.
- **They're natural ranking signals:**
  - boost well-rated or discounted products;
  - push down products that are out of stock;
  - filter by price range.
- **Missing values are stored as `None`, not filled in.** A product without a rating is not a badly rated product, so the ranking stage decides how to treat `None`.
- **Not done yet: scaling.** Prices are very spread out, so a common scale is a decision for the ranking stage (section 4.3).

4. Exploratory data analysis

Decisions taken for the analysis:

- **Scope:** all 28,080 products, with no sampling.
- **Two token views:**
  - *raw* = normalized only, used for counts and word clouds;
  - *processed* = the indexed tokens.

- **Empty fields are counted apart**, not as 0-word texts.
- **Tails are cut only in the plots.** The cut is written on the axis.
- **Ties in the rankings are broken by `pid`**, so the tables can be reproduced.
- All files are in `Part1/eda_results/`.

4.1 Word counts and sentence length

Field (non-empty values)	Empty	Raw words: mean / median / p90 / max	Processed: mean / median / p90 / max
`title`	0	6.5 / 6 / 9 / 36	6.3 / 6 / 8 / 31
`description`	11,150 (39.7%)	50.2 / 39 / 107 / 637	31.2 / 25 / 67 / 579
`product_details` (values)	0	27.3 / 25 / 44 / 206	24.3 / 23 / 38 / 152
merged indexed text	0	66.9 / 46 / 124 / 682	52.1 / 41 / 91 / 623

Source: `word_counts.csv`. Sentences (`sentence_lengths.csv`): 3.3 per description on average (median 2, max 39), and 15.0 words per sentence (median 14, p90 25).

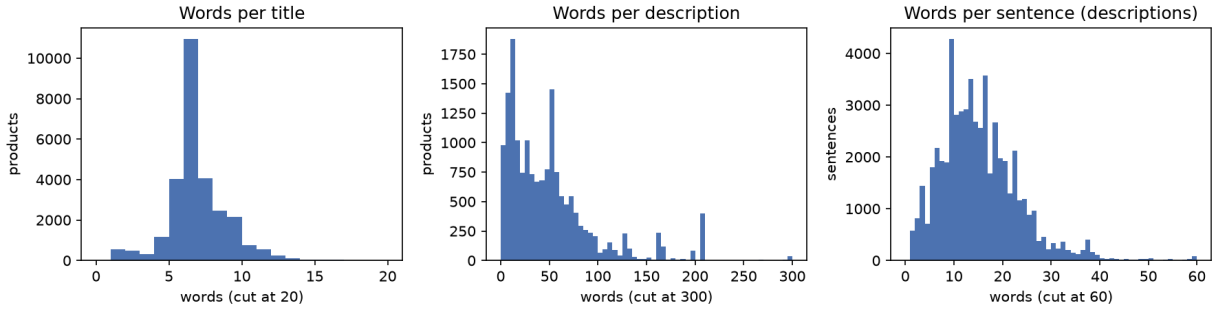


Figure 1: Words per title, per description and per sentence

- **Titles are short keyword templates** (`<pattern> <gender> <neck> <colour> <type>`, e.g. `Printed Men Round Neck Black T-Shirt`).
- **Descriptions are long, skewed and often missing.** The peaks at about 160 and 210 words are template texts copied across products: a care-instructions text in 102 products and a Shoefly polo text in 97 (`repeated_descriptions.csv`).
- **Preprocessing removes 38% of description words but only 3% of title words.** Descriptions are sentences full of stop words, while titles are already keywords.
- **The indexed text ranges from 2 to 682 words**, so the ranking needs length normalization, as BM25 provides.
- **Sentences are split with a regex on `.!?`, not `nltk.sent_tokenize`.** The missing space in `look.Designed` makes a standard splitter see one word.

4.2 Vocabulary, frequent terms and duplicates

Field	Raw terms	Without stop words	Stemmed	Stemmed, seen once
`title`	793	774	743	233
`description`	6,311	6,183	4,704	973
`brand`	424	404	403	42
`product_details`	19,963	19,856	19,366	13,867
merged indexed text	**24,080**	**23,948**	**22,240**	**14,349 (64.5%)**

Source: `vocabulary.csv` (also `sub_category`: 39 terms). The index holds 1,463,607 tokens. The rank-frequency curve follows Zipf's law (`figures/zipf.png`).

Most common terms (stemmed)	Products	Origin
`wear` / `western`	74.7% / 68.1%	`Suitable For = Western Wear` (19,047 products)
`wash`	73.1%	`Fabric Care = Regular Machine Wash` and variants
`cotton`	71.0%	fabric, titles, descriptions
`1`	70.2%	`Pack of = 1` (14,921 products)
`sleev`	64.3%	`Sleeve = Full / Half Sleeve`
`men` / `women`	50.7% / 49.2%	titles, `Ideal For`

Source: `top_terms.csv` (top 25).

- **Long tail of identifiers:**
  - 13,101 of the 13,867 `product_details` terms seen once (94.5%) come from the `Style Code` value, present in 25,763 products (`style_code_terms.csv`).
  - They never match a query, but they inflate the index.
- **Stemming mainly shrinks the description vocabulary** (−24%). Titles and brands are already base-form nouns.
- **Some terms appear in 64–75% of the products, close to stop words of this domain.** They come from boilerplate values.
  - Their IDF is low, so BM25 already gives them little weight, and no domain stop-word list is added. This is reasoning: the effect wasn't measured.
  - `cotton` and `sleeve` are words of query 1, so they'll count little in its score.
- **Heavy duplication** (`duplicates.csv`):
  - Only 5,953 distinct titles: 87.1% of the products share their title with another. The most common is `Slides` (544 products).
  - Only 5,683 distinct descriptions among the 16,930 non-empty ones.
  - Many products will tie on title text alone. This supports indexing `product_details` and using the numeric fields to break ties.

4.3 Numeric fields and rankings

Field	n	Mean	Q1	Median	Q3
`selling_price`	28,078	705.6	390	545	820
`actual_price`	27,303	1,455.5	849	1,199	1,799
`discount` (%)	27,225	50.3	40	53	63
`average_rating`	25,819	3.63	3.2	3.8	4.1

Source: `numeric_summary.csv`.

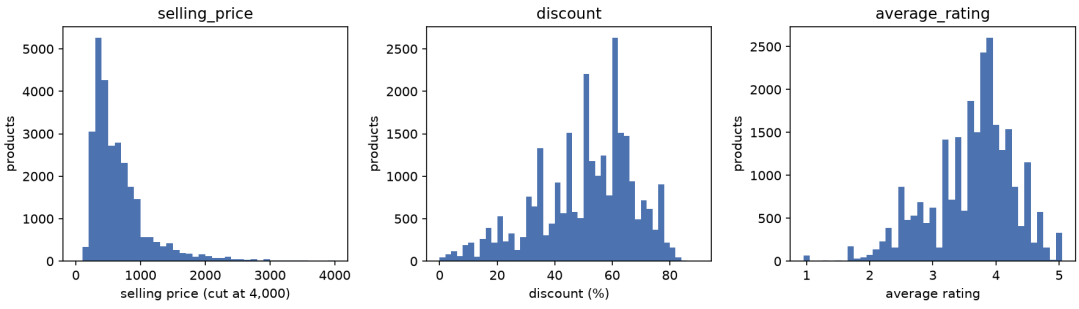


Figure 2: Distribution of selling price, discount and rating

Ranking	Top products (brand as stored, value)
Highest rating	329 products tie at 5.0. The top 10 by `pid` are all briefs and boxers, e.g. Men Brief (Pack of 5) (VIKING.INE).
Most expensive	Solid Women Track Suit (REEB, 7,999); Women Full Sleeve Blazer (True Bl, 7,998); Women Casual Jacket and Men Sweatshirt (Pu, 7,799)
Cheapest	Men Printed Bandana (99); women's socks, pack of 2 (99); ankle socks, packs of 3 (118–120)
Highest discount	Jack Roy women's T-shirts (87%, 86%, 86%); men's T-shirt pack of 2 (yellowvib, 85%)

Source: `top_rating.csv`, `prices.csv`, `top_discount.csv` (top 10 of each).

- **Price is right-skewed** (median 545, max 7,999). A linear 0–1 scaling would squeeze most products together, so a log or rank scale is proposed.
- **High discounts are the norm** (median 53%, half between 40% and 63%), so a discount is a weak signal of a good deal.
- **Ratings cluster between 3.2 and 4.1, and there's no review count.** A 5.0 can't be told apart from a single review, so rating should be a mild boost, not a sort key. This is reasoning.
- The site is Flipkart, so the prices are almost certainly Indian rupees. The dataset doesn't state the currency.

4.4 Brands, sellers, sub-categories and stock

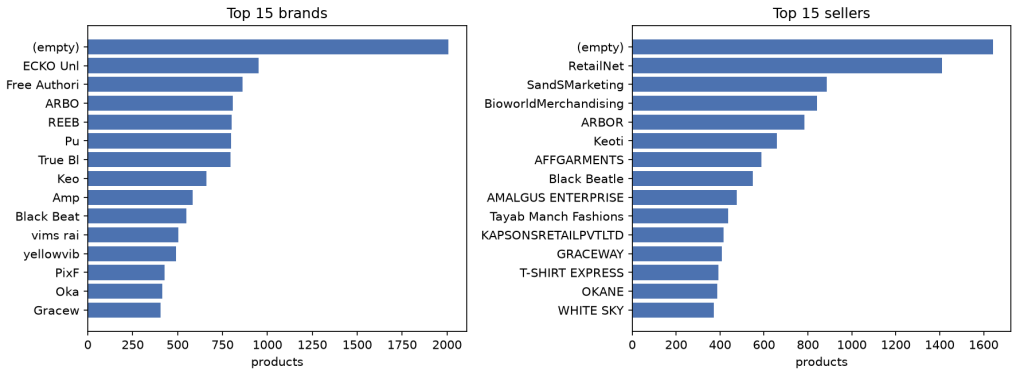


Figure 3: Top 15 brands and sellers (by number of products)

Stored brand	Name in the URL	Products	Brand vs. URL	Products
`ECKO Unl`	`ecko-unltd`	951	cut-off prefix	24,741 (88.1%)
`Free Authori`	`free-authority`	860	empty	2,009 (7.2%)
`REEB`	`reebok`	802	not in the URL	964 (3.4%)
`Pu`	`puma`	798	full match	366 (1.3%)

Source: `truncated_brands.csv`, `brand_truncation.csv`.

- **The stored brand is cut off in 88.1% of the products.** This is the most important finding for retrieval:
  - The full name survives only in the URL.
  - A query for "puma" or "reebok" matches no brand token today, so indexing `brand` (section 2.4) helps little until the name is recovered.
  - The fix isn't applied here, so that the evaluation of section 2 stays valid.
- **An empty seller means out of stock** (`stock_vs_seller.csv`):
  - 1,643 of the 1,644 out-of-stock products have no seller, and every in-stock product has one.

  - So the "largest seller" in the figure is really "no seller", and `seller` adds nothing beyond `out_of_stock`, which supports not indexing it.
- **The catalogue is concentrated** (`top_brands.csv`, `top_sellers.csv`, `sub_categories.csv`):
  - The top 15 brands hold 39.4% of the products, empty brand included.
  - Topwear is 54.5% of the products, so `sub_category` separates other types much better than it separates tops.
- **Ratings differ by brand:** ECKO Unl averages 2.71, ARBO 4.10.
- **`sub_category` has dirty values:**
  - Capitalization variants (`Sunshopping` / `SUNSHOPPING Bags, Wallets & Belts`); lowercasing already merges them in the index.
  - Brand names used as categories (`Uber Urban Clothing and Accessories`).

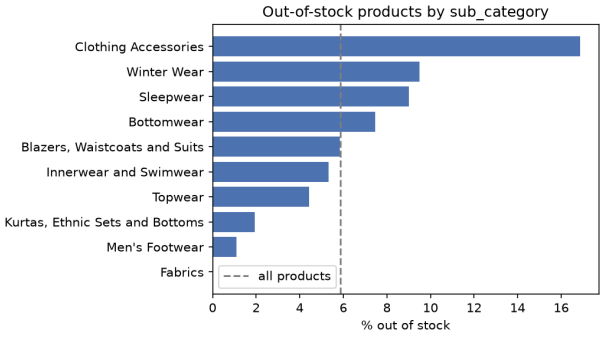


Figure 4: Out-of-stock share by sub_category (at least 100 products)

- **Out of stock** is 5.9% overall, but it ranges from 16.9% (Clothing Accessories) and 9.5% (Winter Wear) to 0% (Fabrics) (`out_of_stock.csv`). These products should be demoted rather than removed, since the user may still want them.

4.5 Word clouds

Decision: the word clouds use raw tokens without stop words and **not stemmed**, because stems such as `typographi` are hard to read.

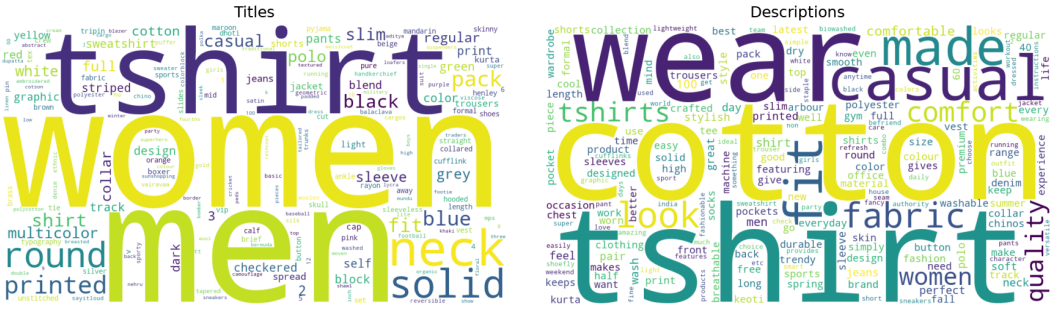


Figure 5: Most frequent words in titles (left) and descriptions (right)

- **Titles:**
  - They're dominated by gender (`men`, `women`), garment (`tshirt`, `shirt`, `polo`, `sweatshirt`, `jeans`) and visual attributes (`printed`, `solid`, `round neck`, colours).
  - These are the words users type, as in the two reference queries.
- **Descriptions:**
  - They're dominated by material and marketing words (`cotton`, `wear`, `casual`, `comfort`, `quality`).
  - They describe material and use more than product type, which explains why they're weaker than `product_details` in section 2.3.

4.6 Entity recognition

Decision: spaCy `en_core_web_sm` over all 16,930 non-empty descriptions (about 1.5 minutes). Titles and `product_details` aren't sentences, so they weren't analysed.

Label	Descriptions with it	Most frequent examples
ORG	51.2%	Printed, KEOTI, Shoefly polo slogan, Sleeve, Tshirt
PERSON	35.3%	Arbour, Casual Wear, Track Pant, Cotton
DATE	25.0%	daily, weekend, all day
PERCENT	21.6%	100%, 60%, 40%
CARDINAL	17.3%	half, 2, 3, two
GPE	15.7%	Cotton, kurta, India

Source: `entities_by_label.csv` (17 labels), `top_entities.csv`.

- **Most labels are wrong for this domain.**
  - The model was trained on news, so capitalized product words become people, places or companies: `Cotton` is tagged as a place, `Casual Wear` as a person.
  - Garments and attributes have no entity type in this model.
- **A few results are useful:**
  - PERCENT gives the fabric composition (60% cotton / 40% polyester).
  - GPE `India` gives the country of origin.
  - Some ORG and PERSON hits are real brands (`KEOTI`, `Arbour`, `Lucky Bird`).
- **Conclusion: NER is not used in preprocessing.**
  - The useful parts are already structured in `brand` and `product_details`.
  - Anonymization (e.g. Presidio) would delete exactly those brand names, which are search terms.

4.7 Implications for the next parts

Query 1 term	Products	Query 2 term	Products
`women`	49.2%	`men`	50.7%
`full`	25.6%	`slim`	20.2%
`sleeve`	64.3%	`jeans`	8.7%
`sweatshirt`	5.0%	`blue`	25.8%
`cotton`	71.0%		

Source: `query_term_df.csv` (share of products whose indexed text contains the term).

- **Each query has one rare, decisive term** (`sweatshirt`, `jeans`). With IDF weighting it picks the candidates, and the common terms order them. This matches section 2.4: the labels separate products by sleeve and gender, not by type.
- **Proposed for the indexing stage** (not applied, to keep section 2 valid):
  1. Recover the full brand from the URL path.
  2. Drop the `Style Code` value.
  3. Demote, not filter, out-of-stock products, and use the numeric fields to break ties.
  4. Use a log or rank scale for price, and rating as a mild boost.